THE PULLBACK OF A SMALL ETALE SHEAF TO THE FPPF TOPOLOGY IS DESCRIBED BY THE SECTIONS OF ITS SMALL PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the etale statement and the fppf sheaf condition, both read off the etale description lemma.

Editorial note. The label lemma-describe-pullback-pi-fppf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

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Allowed standard prerequisites (editorial).

- The description of the pullback in the etale topology. Same chapter, the lemma identifying pi_S^{-1}F with the rule (f : X -> S) |-> Gamma(X, f_small^{-1}F), which also proves the fppf sheaf property. Supplies both the etale part of the statement and the fppf sheaf condition.

Source reference. Original reference 09XN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4960–4960. Prerequisite pointer invoked at proof line(s) 21469; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation as above. Let $\mathcal{F}$ be a sheaf on $S_{étale}$. The rule*

$$(Sch/S)_{fppf} \longrightarrow Sets, \quad (f : X \to S) \longmapsto \Gamma(X, f_{small}^{-1}\mathcal{F})$$

is a sheaf and a fortiori a sheaf on $(Sch/S)_{étale}$. In fact this sheaf is equal to $\pi_S^{-1}\mathcal{F}$ on $(Sch/S)_{étale}$ and $\epsilon_S^{-1}\pi_S^{-1}\mathcal{F}$ on $(Sch/S)_{fppf}$.

Proof. The statement about the étale topology is the content of Lemma 09XN. To finish the proof it suffices to show that $\pi_S^{-1}\mathcal{F}$ is a sheaf for the fppf topology. This is shown in Lemma 09XN as well. □

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.